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# Interface mechanics of liquid crystal polymer nanocomposites with high concentrations of MWCNTs

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ABSTRACT

Compositing a liquid crystal polymer (LCP) with highly oriented multi-walled carbon nanotubes (MWCNTs) at non-dilute filler concentrations has attracted considerable attention due to its function as alignment layer as well as mechanical reinforcement. In this study, the structural and mechanical features of the LCP–MWCNT interface were investigated by all-atom molecular dynamics (MD) simulations. The results suggested that the MWCNT filler co-oriented with the liquid crystalline matrix enhances the crystallinity of the LCP molecules, especially in the interfacial area. The aromatic rings of the LCP components exhibited a repeating structural pattern because of the different equilibrium distances depending on the type of π-π stacking. The effect of the morphological change of LCPs in the interfacial region on the load transfer capability was also evaluated in terms of filler concentrations. In particular, tight coupling between the liquid crystal components and efficient load transfer to the outermost nanotube shell were first identified. These features account for the enhanced mechanical stiffness in the transverse loading direction with increasing volume fraction of the MWCNTs, as confirmed experimentally.

## 1. Introduction

Liquid crystalline polymers (LCPs) are attractive stimuli-responsive materials that exhibit mechanical anisotropy. When liquid crystal monomers polymerize to form main- or side-chain segments, the mesogenic groups in liquid crystalline states maintain a long-range orientational order. Such ordering of mesogens can be selectively disrupted by heating or light irradiation at a specific wavelength to impose large mechanical strain throughout the polymer system [1,2]. Thus, by precisely controlling the magnitude of deformation responding to external stimuli and the resulting actuation force, LCPs can be applied as a driving component of soft robotic systems [3–6].

Recently, to improve mechanical properties and impart newer functionalities, efforts have been directed at the development of LCP nanocomposites, wherein LCPs and multi-walled carbon nanotubes (MWCNTs) coexist. It is known that well-oriented carbon nanotube structures assist the orientation of LCPs to align with the nanotubes during the mixing process [7–10]. Owing to the excellent mechanical properties of MWCNTs, the modulus of elasticity and mechanical strength of LCP nanocomposites are superior than those of neat LCP without any reinforcement [9,10]. In addition, due to their high optical absorption in the visible/near-infrared region, MWCNTs can serve as an ultrafast heat source under light irradiation, which can further increase the versatility of LCP-based systems [1,9].

Two crucial factors that govern the mechanical properties of LCPs are the molecular-scale orientation state of the mesogenic groups and their mutual binding force. In the case of LCP nanocomposites, load transfer characteristics between the matrix and reinforcing agent, and a distinct morphological feature of the interfacial region additionally involved in the properties. Although early studies have attempted to elucidate the nature of interfaces in polymer/MWCNT nanocomposites [11,12], only a few studies have investigated the mutual arrangement characteristics with MWCNTs when the matrix is composed of LCPs. Moreover, most of the LCP/MWCNT nanocomposites have a very high filler concentration (>10 vol%) because they are fabricated by infiltrating the liquid crystals into dense MWCNT layers extracted from an MWCNT forest [9,10]. This makes analytical composite models constituted at dilute filler concentration conditions be unsuitable for the system. Methodologies for predicting the load transfer characteristics and the associated internal stress distribution inside LCP/MWCNT nanocomposites during mechanical loading have not been fully explored either.

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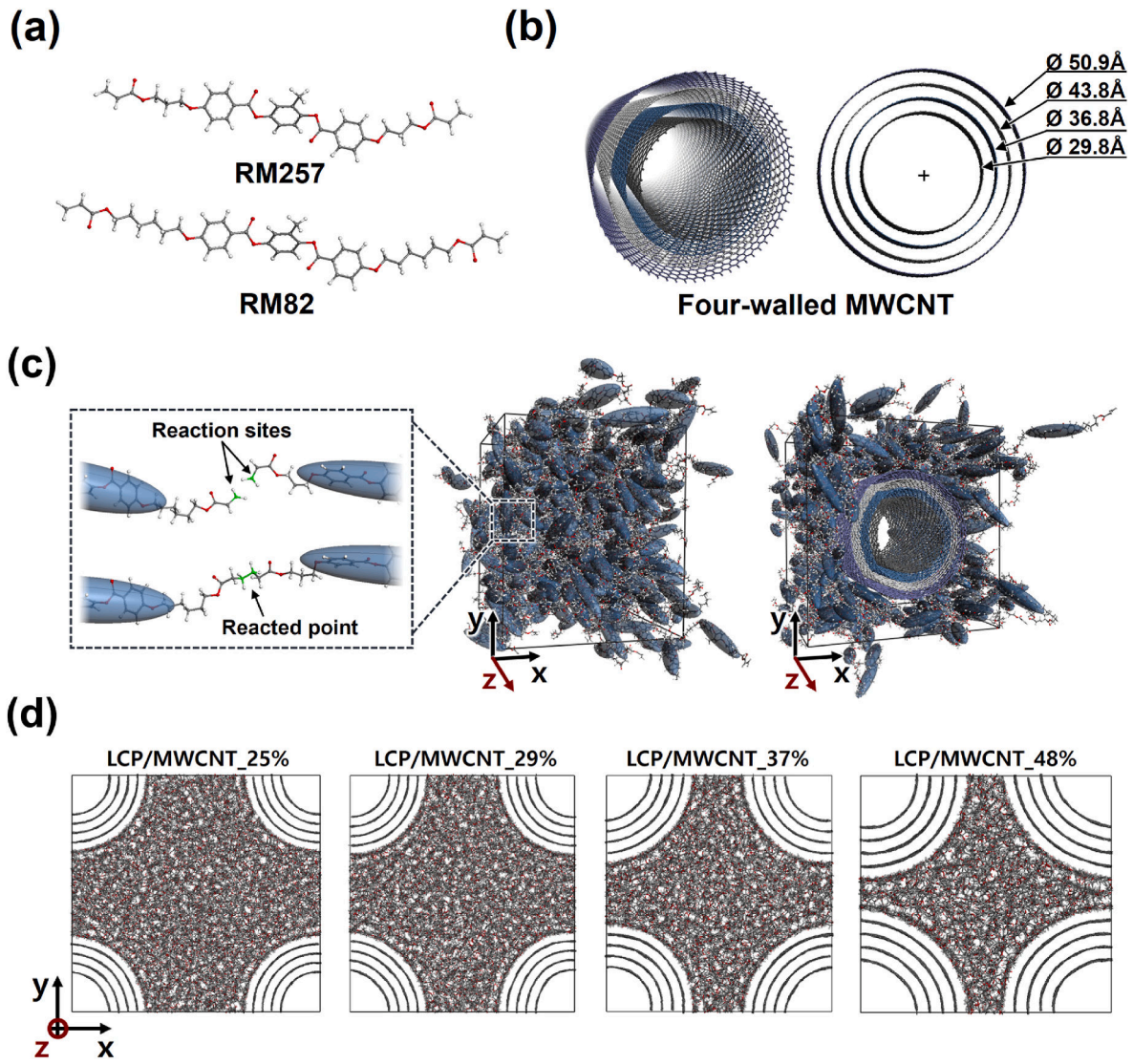



**Fig. 1.** Molecular configurations of the modeled neat LCP and LCP/MWCNT nanocomposite systems. (a) Molecular structure of the liquid crystal monomers and (b) multi-walled carbon nanotube (MWCNT) with four walls. (c) Snapshots of the nematic LCP and LCP/MWCNT_29% models after equilibration. The mesogenic groups in the liquid crystal monomers are emphasized as ellipsoids. (d) Front-view of LCP/MWCNT nanocomposites with different MWCNT volume fractions. To emphasize the polymer conformation between the MWCNTs, all unit cell sizes were normalized and the center of the nanotube was placed at the boundary of the periodic unit cell.

Considering the aforementioned factors, we examined the mechanical behavior of LCP nanocomposites with a high concentration of MWCNTs using all-atom molecular dynamics (MD) simulations. By evaluating the orientation and alignment properties of the LCPs in the vicinity of the MWCNT surface, the influence of oriented nanotubes on the director field of the matrix was investigated *in silico*. The strain energy density assigned to each molecular component of the nanocomposite was quantified during mechanical loading. Changes in the mechanical anisotropy and intermolecular bonding properties of the LCP/MWCNT nanocomposites were also evaluated according to the MWCNT volume fraction. Finally, the theoretical model was validated experimentally using planar aligned LCP/MWCNT composites.

So far, the microscopic mechanics of LCP molecules in the interaction with MWCNT sheets have not been explored sufficiently. The main objective of this study is to unlock the atomistic details of the LCP-MWCNT interface from the mechanical and structural points of view. The interplay between MWCNTs and LCPs was systematically analyzed under conditions of high MWCNT volume fraction. In particular, the effect of each component constituting the nanocomposites on the mechanical properties in the interfacial region was described as continuum Cauchy stress. Therefore, the present study provides fundamental insights into the mechanical role of the interface in LCP/MWCNT nanocomposites, potentially broadening the scope of mechanical design for LCP-based smart structures.

## 2. Molecular modeling of LCP and LCP/MWCNT nanocomposites

Fig. 1 shows the LCP and LCP/MWCNT nanocomposite models constructed in the MD simulation environment. Two types of liquid crystal molecules, 1,4-bis-[4-(3-acryloyloxypropyloxy) benzoyloxy]-2-methylbenzene (RM257) and 1,4-bis-[4-(6-acryloyloxyhexyloxy) benzoyloxy]-2-methylbenzene (RM82), were selected to construct the LCP matrix. Further, an MWCNT consisting of a set of zigzag (65,0), (56,0), (47,0), and (38,0) tubes that were concentrically arranged in the z-direction was prepared in the current study. All tubes were designed to have a constant length of 68.2 Å, but they behave as infinite cylinders during MD runs as the end atoms of each tube were interconnected across the periodic boundary. The diameter of the tubes was from 29.8

**Table 1**
LCP and LCP/MWCNT nanocomposite systems used in MD simulations.

| Model | Crosslinking ratio (%) | Unit cell length (Å$^3$) | Nearest distance between MWCNTs (Å) | MWCNT volume fraction (%) /weight fraction (%) | Density (g/cc) |
|---|---|---|---|---|---|
| Neat LCP | 62.6 | 62.6 × 73.1 × 68.9 | – | – | 1.16 |
| LCP/MWCNT_25% | 62.4 | 93.2 × 88.6 × 67.0 | 40.0 | 24.6/35.8 | 1.33 |
| LCP/MWCNT_29% | 60.8 | 81.4 × 85.5 × 67.0 | 32.6 | 29.2/41.7 | 1.35 |
| LCP/MWCNT_37% | 61.5 | 72.9 × 76.1 × 67.0 | 23.6 | 36.7/50.5 | 1.40 |
| LCP/MWCNT_48% | 60.0 | 67.2 × 63.4 × 67.0 | 14.3 | 47.7/62.5 | 1.47 |

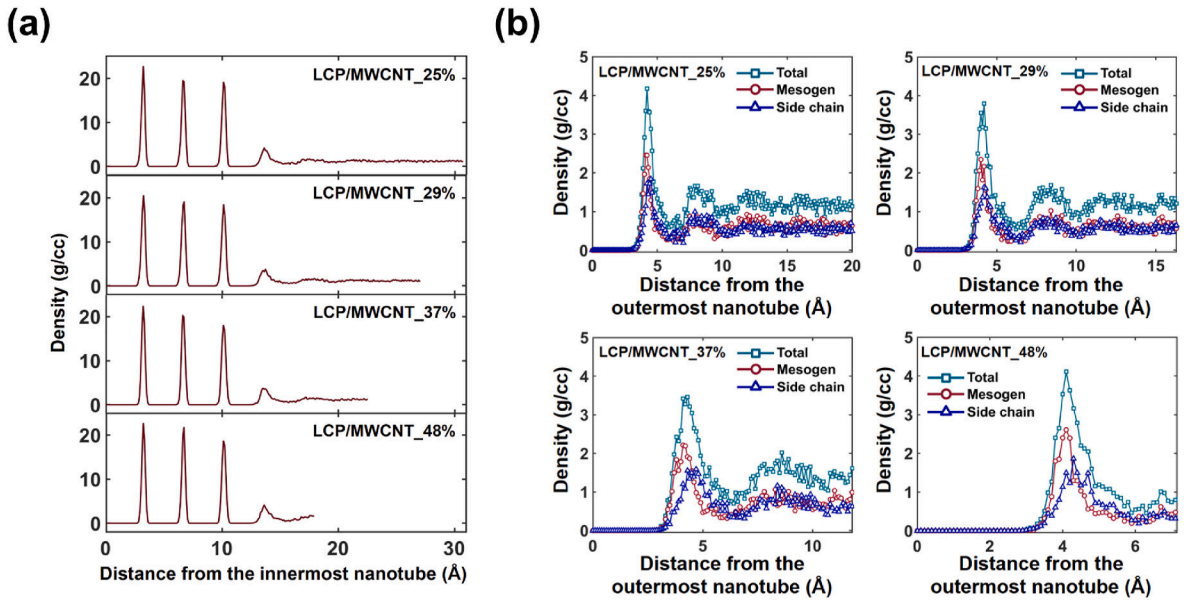



**Fig. 2.** Density distributions of the cylindrical shells of LCP/MWCNT nanocomposites. Density profiles from the (a) innermost nanotube surface and (b) outermost nanotube surface.

to 50.9 Å, which is close to the experimental conditions [9]. The details of the construction of the LCP and LCP/MWCNT nanocomposites are described in the simulation model section (also refer Fig. S1, **S2**, **S3**, and Table S1) in the Supplementary Data. The validation of the structural properties of the models prepared can be founded in Fig. S4 and Table S2 in the Supplementary Data. After equilibration, the MWCNTs have a different volume fraction in the range of 24.6–47.7 vol%, respectively. The structural properties of the models considered are also presented in Table 1.

## 3. Results and discussion

### 3.1. Structural features of the LCP/MWCNT nanocomposites

To investigate the structural characteristics of the LCP under the influence of the MWCNT filler, the inner space of the unit cell was divided into several sets of cylindrical shells with a thickness of 0.1 Å, and the density of each region was calculated. The density profiles starting from the innermost nanotube surface are shown in Fig. 2(a). After the formation of three peaks of 20 g/cc by evenly spaced nanotubes, a densely adsorbed LCP layer appeared on the outermost nanotube surface. Then, the profiles in the region of the LCP matrix gradually converged to the bulk value of 1.16 g/cc as the distance from the MWCNT increased. During the convergence of density, the density distribution of the entire LCP matrix did not remain steady, but fluctuated significantly, which is a well-known feature that appears at the interface of polymer nanocomposites [13–16]. These fluctuations were eliminated at a distance of ~15 Å from the outermost nanotube surface. Meanwhile, the peak density of the adsorbed LCP layer remained constant at ~4 g/cc, regardless of the MWCNT concentration.

A focused view of the density distribution in the LCP region is presented in Fig. 2(b). To further examine how each LCP component contributes to the overall density distribution of the LCP matrix, the density profiles of the mesogenic units and side chains were obtained independently. As shown, the density profiles of both the mesogen and the side chain exhibit a considerable correlation and highly oscillate depending on the distance from the nanotube surface. Remarkably, the peak value of the mesogen density is higher than that of the side chain, and the corresponding position is also closer to the MWCNT surface. These results can be attributed to effective π-π interactions between the MWCNT surface and the aromatic rings in the mesogen.

For equilibrated LCP/MWCNT nanocomposite models, the distribution of the orientation angle ($\theta$) between the MWCNT surface and aromatic ring of the mesogen was derived, as shown in Fig. 3(a). Each value was defined as the angle between the normal vector of the plane formed by the aromatic ring ($\overrightarrow{N_1}$) and the normal vector of the nanotube surface placed at the shortest distance from the center of the aromatic ring ($\overrightarrow{N_2}$). Meanwhile, the LCP matrix region of the nanocomposite was divided into a set of cylindrical shells with a thickness of 0.1 Å surrounding the nanotube, and the orientation angles in each shell region were averaged. In terms of π-π stacking between the aromatic groups, the rings can theoretically have various configurations such as sandwich (the rings arranged parallelly and concentrically), parallel-displaced (the rings arranged parallelly, but non-concentrically), and T-shaped (one ring perpendicular to another), etc [17,18]. We simply classified the mesogen morphology around the MWCNT filler into two representative configurations: parallel alignment ($\theta = 0°$) and T-shaped ($\theta = 90°$).

The orientation angle distribution from the outermost nanotube surface and the morphologies of the aromatic rings around the MWCNT

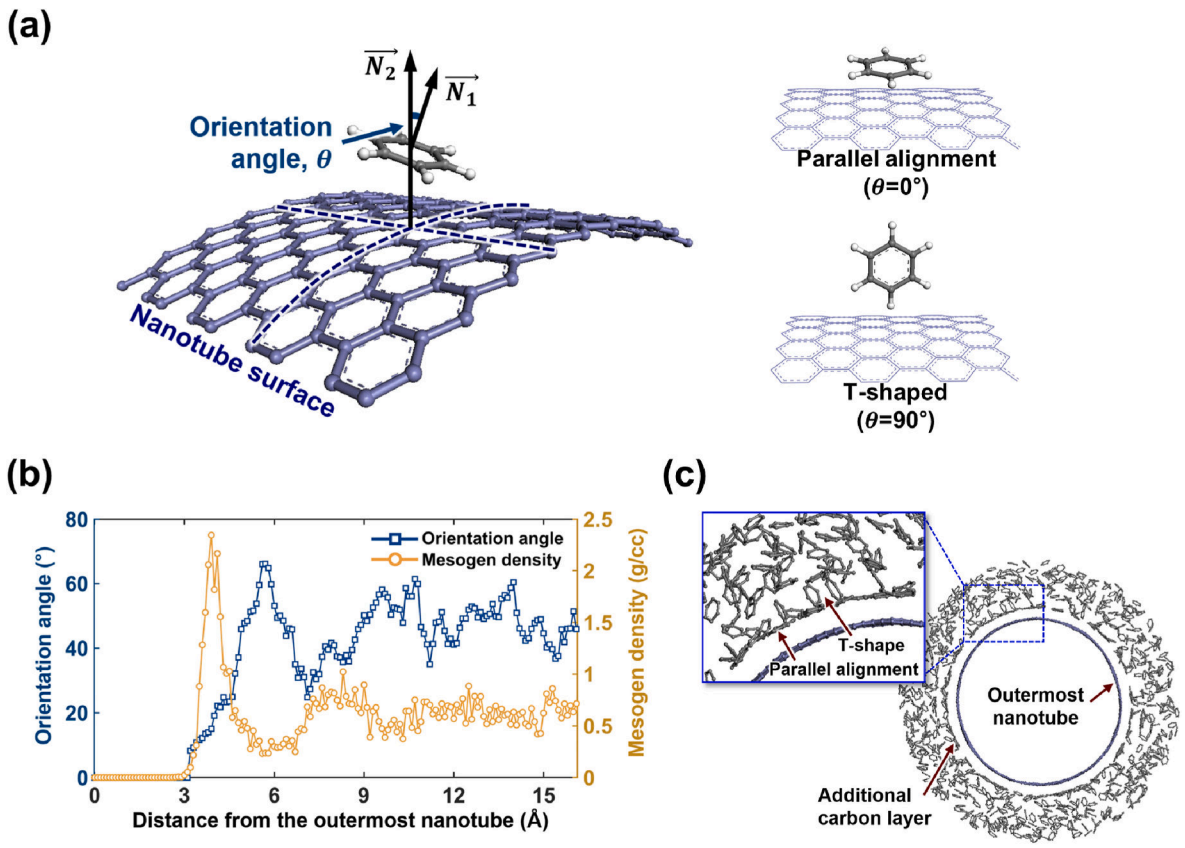



**Fig. 3.** Distribution of the aromatic ring conformations around the MWCNT surface. (a) Schematic of the measurement of the orientation angle and two representative conformations of the aromatic ring. (b) Orientation angle distribution and mesogen density profile of the LCP/MWCNT_29% nanocomposite model in the radial direction. The orientation angle data was smoothed using a five-point moving average for clarity. (c) Snapshot of the aromatic rings around the nanotube surface in the LCP/MWCNT_29% nanocomposite model. All atoms except the outermost nanotube and carbon rings of the mesogen units are hidden for clarity.

are presented in Fig. 3(b) and **(c)**, respectively. The space in the immediate vicinity of the outermost nanotube surface was predominantly occupied by the parallel aligned mesogen units, as reported earlier for polymers with aromatic rings in the backbone [19–22]. However, in the region of 4.5–7 Å, the mesogen units rather preferred to align perpendicular to the nanotube after the parallel alignment layer. The reason why the structural characteristics are clearly classified for each radius away from the MWCNT surface is that the equilibrium distance is different for each alignment type. For parallel alignment, the equilibrium distance between the ring centroids is about 3.4 Å (in the parallel-displaced configuration, only for vertical offset) or 3.9 Å (in the sandwich configuration). In contrast, the equilibrium state of the T-shaped configuration requires a longer distance of 5.0 Å [23,24]. Accordingly, the two types of layer structures alternate with increasing distance from the MWCNT surface, while the average angle gradually converges to 45°, indicating a random distribution. Note that the fluctuation of the mesogen configuration disappears completely at a distance of 15 Å from the MWCNT surface, which is farther than the noncovalent interaction cutoff of 10 Å applied in the current study. The results indicate that the mesogen units arranged parallel to the nanotube surface act as an additional aromatic carbon layer. Since the aligned mesogen units are concentric to the MWCNT filler, another interlayer interaction between the mesogenic groups occurs over a longer range.

Note that both π-π stacking of aromatic rings on the nanotube surface exhibits markedly higher binding energy (7.59 kcal/mol for parallel alignment and 4.00 kcal/mol for T-shaped alignment) than the van der Waals energy of ordinary polymeric constituents (less than 1 kcal/mol) [18,33]. In addition, from a structural point of view, LCP molecules are arranged alternately in parallel and T-shape along the radial direction of MWCNTs. This results in a highly packed interfacial microstructure for effective load transfer. Consequently, these layered polymer structures play a key role in determining the stress transfer capability from the matrix to the filler under mechanical loadings, which will be further discussed in section 3.2.

Meanwhile, the angle distribution results described above strongly correlate with the radial density distribution profile presented in Fig. 3 (b). The density profile of the mesogenic units has an inversely proportional dependence on the orientation angle distribution, indicating that microstructures composed of parallelly aligned units appear predominantly, whereas T-shaped microstructures are scarce. As a majority

**Table 2**
Orientational order of liquid crystal components according to the models used in this study.

| | Neat LCP | LCP/MWCNT _25% | LCP/MWCNT _29% | LCP/MWCNT _37% | LCP/MWCNT _47% | Ref. [25] | Ref. [26] |
|---|---|---|---|---|---|---|---|
| Liquid crystals | RM257 & RM82 | | | | | RM257 | RM82 |
| $S_m$ | 0.625 | 0.613 | 0.609 | 0.606 | 0.565 | – | – |
| $S_s$ | 0.318 | 0.314 | 0.280 | 0.339 | 0.260 | | |
| Exp. | – | | | | | 0.538 | 0.545 |

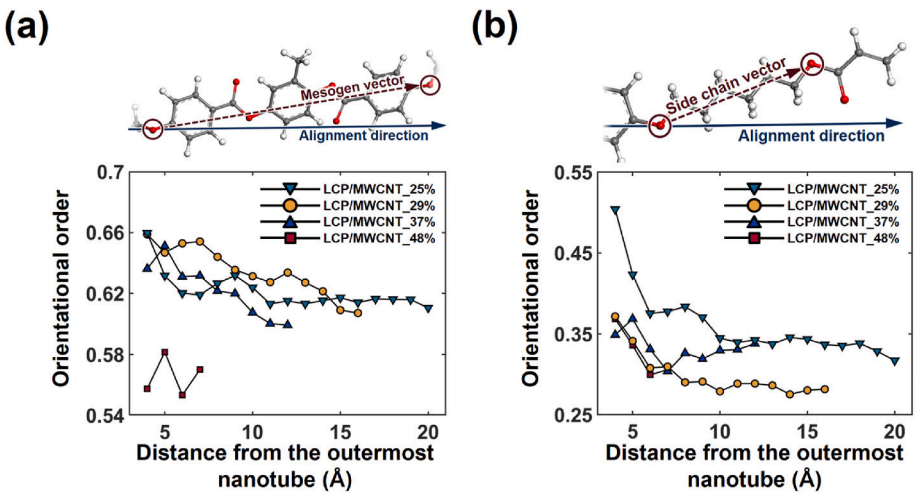



**Fig. 4.** Radial distribution of the orientational order parameter for the (a) mesogens and (b) side chains. The method for measuring the angle formed by each molecular component is illustrated at the top.

of mesogens adopted the parallel alignment state across the entire surface of the MWCNT, only a few T-shaped configurations were found in the outer area after the completion of parallel alignment (see Fig. S5 in the Supplementary Data).

To estimate the change in the directionality of the LCP matrix due to the insertion of the aligned MWCNT filler, the alignment distribution of the liquid crystal molecules was assessed. The rigid mesogens and flexible side chains were considered separately to clarify the features of each component in the system. The vector penetrating the oxygen atoms at both ends of each component was defined and the corresponding angle formed with the alignment direction (i.e., z-axis) was estimated. The estimated value was quantified through the orientation order parameter ($S_i$) defined as follows:

$$S_i = \langle \frac{3\ cos^2\theta_i - 1}{2} \rangle,\ i = m \text{ or } s \tag{1}$$

where the term inside brackets, $< >$, indicates the average value for a specific component of the LCP matrix, and the subscripts $m$ and $s$ refer to the mesogen and side chain, respectively.

Table 2 shows the orientational order of LCP matrix predicted by the MD model, along with previously reported experimental data. The orientaional order of liquid crystal molecules reported in Refs. [25,26] is the results of wide angle X-ray scattering (WAXS) experiments. Note that the diffraction pattern obtained from WAXS is critically sensitive to the arrangement of the mesogenic group, thus it is frequently used to represent the properties of mesogen. Accordingly, it is believed that the arrangement of the mesogens in the neat LCP model shows good consistency with the experimental observations, validating the present computational approach. Meanwhile, the alignment properties of the components in the LCP/MWCNT nanocomposite models were almost identical, despite a high loading of MWCNTs. This result is interesting because at a high volume fraction, the MWCNT filler spatially interferes with the molecular arrangement of the liquid crystalline material. That is, the LCP molecules not only organize among themselves but also show high directionality with the MWCNT arrangement when they first encounter the MWCNT surface.

The radial distributions of the orientational order parameter of each component in the LCP/MWCNT nanocomposite models are shown in Fig. 4. Notably, the profiles of both the mesogen and the side chain have high directionality, especially in the vicinity of the MWCNT surface. When liquid crystal molecules are placed on the surface of MWCNTs, a strong anchoring effect is induced due to the π-π stacking energy discussed above. Meanwhile, from a structural viewpoint, MWCNTs have a very high aspect ratio, while mesogenic units can be considered as rigid ellipsoids with a high bending strength. If the mesogenic core encounters the MWCNT diagonally or perpendicularly, the high curvature of the nanotube prevents the core from tightly wrapping the surface. From the viewpoint of minimizing the deformation energy inside the composite component, therefore, it is most desirable for the mesogenic core to be aligned with the MWCNT. Also, the contact area increases as the off-axis angle decreases between the mesogen and MWCNT, forming a stronger non-covalent interaction force. This indicates that the LCP matrix spontaneously forms a parallel arrangement with the MWCNT when the LCP molecules approach the pre-aligned MWCNTs.

Similarly, the stable state of the side chain is also achieved by arranging it on the nanotube surface. A linear structure of the side chain can exhibit two different conformations, lying and standing structures, depending on the angle with the nanotube [34]. From the energy point of view, the adsorption energy of the lying structure formed by aligning with the nanotube is stronger than the one of the standing structure formed by wrapping the nanotube [35]. Therefore, it is plausible that the side chains also prefer to be arranged parallel to the nanotubes. The structural feature covalently bonded to the ends of the nematic mesogens also influences the directionality of the side chains. The effect of the MWCNT on molecular alignment gradually decreases with increasing distance from the outermost nanotube, and the orientational order finally reaches that of the bulk LCP.

### 3.2. Mechanical properties of the LCP and LCP/MWCNT nanocomposites

The mechanical properties of the equilibrated models in the elastic region were estimated through tensile loading simulation. A uniaxial deformation force with a constant strain rate of $10^6$/s was applied until the total strain reached 0.1%. During elongation, the atomic stress tensor was calculated based on the virial formula [27]. The uniaxial tensile loading simulations were conducted under the NVT ensemble with a low temperature of 0.1 K where the thermal vibration of the constituent atoms was eliminated. Therefore, the virial stress is equivalent to mechanical Cauchy stress [28]. Meanwhile, the stiffness components given in Voigt notation ($C_{ij}$) were derived as follows:

$$\sigma_i = C_{ij}\varepsilon_j\ (i,\ j = 1,\ \ \ldots,\ 6), \tag{2}$$

where $\varepsilon_j$ is the strain applied to the system. As both the LCP and the LCP/MWCNT are transversely isotropic, the stiffness components perpendicular to the alignment direction were averaged as follows:

$$C_{11} = \frac{(C_{xx} + C_{yy})}{2} \quad \text{and} \quad C_{33} = C_{zz} \tag{3}$$

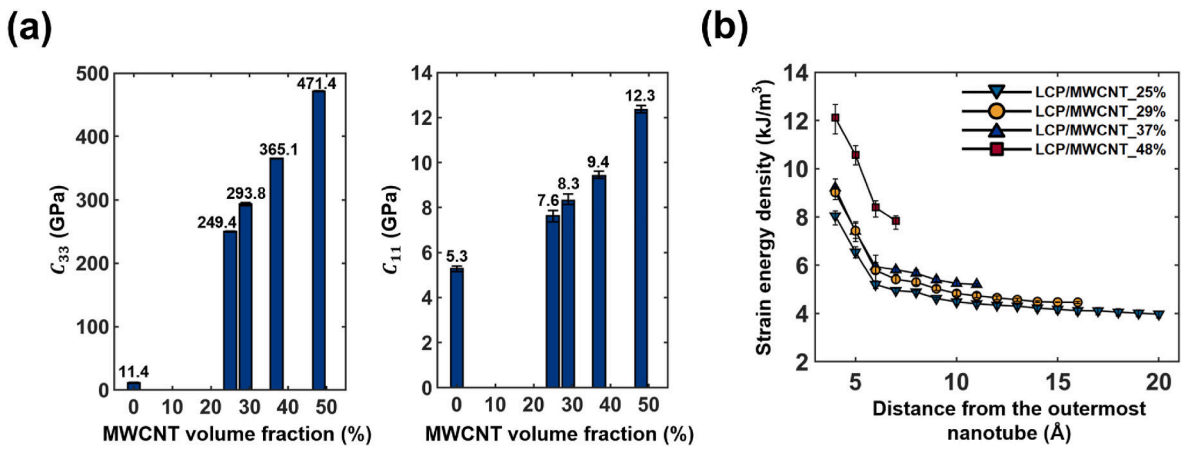



**Fig. 5.** Mechanical properties of the LCP and LCP/MWCNT nanocomposites. (a) Elastic stiffness components of the LCP and LCP/MWCNT nanocomposites in axial (left) and transverse (right) directions. An MWCNT volume fraction of 0% represents the results for the neat LCP sample. (b) Strain energy distributions of the LCP matrices in the LCP/MWCNT nanocomposites under transverse loading. Error bar indicates the minimum/maximum range.

All the mechanical tests were performed in triplicate and the results are averaged.

Fig. 5(a) shows the stiffness components calculated in the axial (i.e., z-axis) and transverse directions (i.e., x- and y-axes) according to the volume fraction of the MWCNT. The neat LCP model, which represents an MWCNT volume fraction of 0%, showed anisotropic elastic properties. Meanwhile, the insertion of the MWCNT into the LCP matrix improved the mechanical properties of the system in all directions. However, the degree of enhancement in each direction varied significantly. Under axial loading, the elongation of the LCP/MWCNT unit cell directly stretches the infinite MWCNT in the longitudinal direction. Accordingly, the elastic stiffness increased linearly by up to 40 times compared to that of the neat material, depending on the MWCNT volume fraction. A more interesting aspect of this study is the strengthening effect of the MWCNT under transverse loading. It has been well recognized that the load transfer efficiency against transverse deformation is quite poor because of weak and slippery interfaces between the polymer matrix and carbon nanotube [29,30]. Nevertheless, the liquid crystal chains in the vicinity of the MWCNT surface achieve considerable stability from a structural point of view because of their arrangement owing to intermolecular interactions such as the stacking of the aromatic rings. Therefore, the elastic stiffness under transverse loading also improved by up to 2.3 times compared to that of the neat LCP sample. Another remarkable observation is that the transverse stiffness changed exponentially with increasing MWCNT volume fraction. The results suggest that the concentration of MWCNTs is a key factor that determines the load transfer characteristics at the interface between the LCP matrix and MWCNT filler.

To further investigate the structural characteristics of the LCP matrix around the MWCNT under transverse loading, the strain energy density distribution was calculated. Starting from the outermost nanotube, the LCP atoms within a specified cutoff distance were individually grouped into cylindrical rings. The cutoff distance was initially set as 4 Å and increased incrementally by 1 Å until the unit cell boundary was reached. Then, the strain energy density ($u$) of each group was computed as follows:

$$u = \int_0^{\varepsilon_0} \sigma d\varepsilon, \tag{4}$$

where $\sigma$ and $\varepsilon$ are the stress and strain applied to each cylindrical group, and $\varepsilon_0$ indicates the total strain of 0.1%. The transverse loading simulation was carried out for the x- and y-axes, respectively, and the results from the two were averaged.

Fig. 5(b) presents the strain energy density distributions of the LCP/MWCNT nanocomposites under a transverse load of 0.1%. Since all the LCP/MWCNT nanocomposite models were elongated with a constant strain amplitude, each point in the distribution represents the relative deformability of the local polymer conformation [31,32]. As shown, the strain energy density profiles of all models considered are not uniform, but are distributed with respect to the distance from nanotube surface. In particular, a distinctly high value was observed in the nearest vicinity of the MWCNT surface, indicating significant slippage of the interfacial LCP in response to the external load. It is worth noting that the crystallinity of the LCP in the corresponding region was considerably enhanced (refer to Figs. 3 and 4). This improves the structural rigidity and makes the liquid crystal components couple to each other tightly. Under transverse loading, therefore, the external force drags the well-ordered liquid crystal molecules as a single unit against the weak anchoring force produced by the MWCNTs, causing enlarged deformation at the interface. The effect gradually decreases as the distance from the nanotube increases, and the crystallinity of the LCP returns to that of the bulk state. The slope of each profile flattens at a distance of 6 Å, which is also consistent with the thickness of the first adsorbed layer of the LCP matrix (see Fig. 2(b)). That is, as opposed to the first adsorbed layer with high crystallinity, the deformability of the far-field layers was relatively suppressed. Meanwhile, the overall energy density profile rose with the increase in the MWCNT volume fraction, representing the improvement in the mechanical properties of LCP domain in the transverse direction.

To evaluate the influence of the MWCNT volume fraction on the load transfer characteristics of the nanocomposite, the effective strain energy density assigned to each molecular component was analyzed. The components of the neat LCP and LCP/MWCNT nanocomposites were classified into three groups, viz., mesogen, side chain, and nanotube layer, as described in Fig. 6(a). For calculating the virial stress under mechanical loading, the volume occupied by each phase (i.e., LCP matrix or MWCNT filler) was substituted for the volume variable. The method allows a quantitative comparison of the mechanical contribution of the components between the nanocomposites with different MWCNT volume fractions.

Fig. 6(b) shows the effective strain energy density of the neat LCP model. The strain energy contributed by covalent and noncovalent bonds was also calculated separately to estimate intra- and intermolecular interactions precisely. The results revealed mechanical anisotropy of both components. Under axial loading, the mesogenic group experiences more external load than the side chain group because of the difference in their structural rigidities. As the highly oriented diacrylate liquid crystal members were loaded along the covalent path, a majority of the strain energy was allocated to the covalent bonds in both

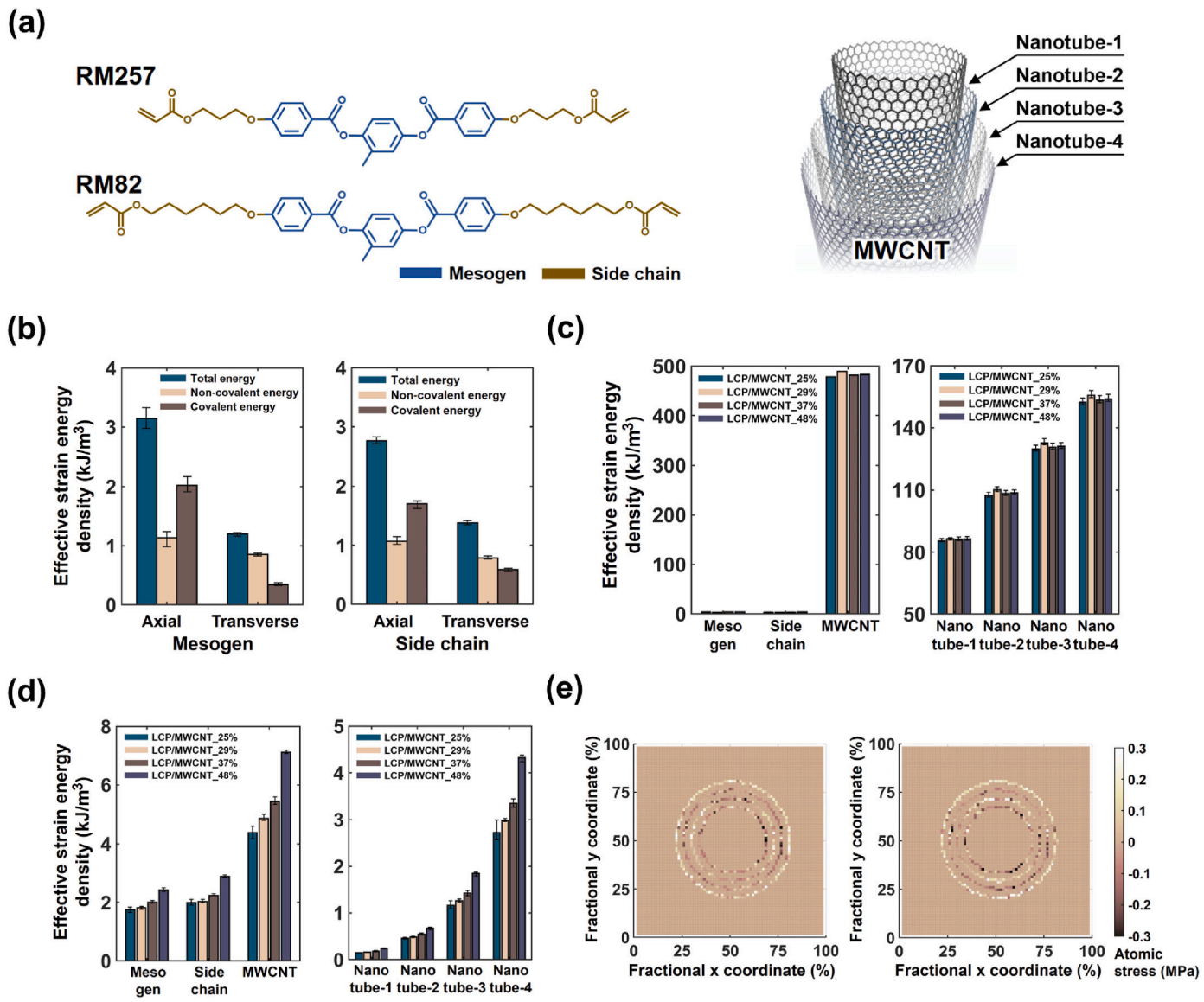



**Fig. 6.** Deformation energy distributions for each molecular component under a uniaxial tensile strain of 0.1%. (a) Classification of the molecular components of the LCP and MWCNT. (b) Effective strain energy density of the neat LCP model. (c) Results for the LCP/MWCNT nanocomposite models during axial loading and (d) transverse loading. (e) Stress contours of the MWCNT in the LCP/MWCNT_29% nanocomposite under a transversal tensile strain of 0.1%. The unit cell was divided into 75 × 75 elements along the x- and y-axes, and all the atomic stresses generated in each cubic region were cumulated. The mechanical load was applied along the x-axis (left) and y-axis (right), respectively. Error bar indicates the minimum/maximum range.

groups (~60% of the total energy). In contrast, under transverse strain, the contribution of the noncovalent strain energy increased compared to the case of the axial strain. In the analysis according to the molecular component, the ratio of the strain energy density of the side chain to the total strain energy increased significantly. This result suggests that the mechanical properties of the entire system during transverse loading are rather predominantly determined by the deformability of the flexible side chains and the associated noncovalent intermolecular forces.

The results for the LCP/MWCNT nanocomposite models under axial deformation are shown in Fig. 6(c). As expected, the MWCNT filler accommodates the entire applied load, and the magnitude of strain energy density remained almost unchanged regardless of the MWCNT volume fraction. In this case, the strain energy of each nanotube group is also independent of the volume occupied by the MWCNT. Note that the lower strain energy density in the inner nanotube is due to the decrease in the number of carbon atoms in the nanotube, and the substantial contribution of each constituent atom is almost constant (the difference between adjacent nanotube groups is within 2%; refer to Fig. S6 in the Supplementary Data).

In contrast, as depicted in Fig. 6(d), the deformation energy induced by transverse loading was optimally distributed between the LCP matrix and MWCNT filler, and the values for all components exhibit an increasing tendency with the increase in the volume fraction of the MWCNT. In addition, in contrast with the response to axial loading, an exponential distribution of the strain energy according to the nanotube layer is also noticed. Such distribution of the load for each nanotube group could also be found in the atomic stress calculation results for each spatial grid element, and the representative stress map is shown in Fig. 6(e).

The exponential enhancement of the mechanical properties of the LCP/MWCNT nanocomposites according to the MWCNT concentration under transverse loading suggests the following mechanism: As discussed in Figs. 3 and 4, the highly oriented MWCNTs assist the alignment of the neighboring liquid crystal molecules and improve their crystallinity. Next, as this ensures a high binding strength between the liquid crystal members over a long distance, the LCPs influenced by MWCNTs

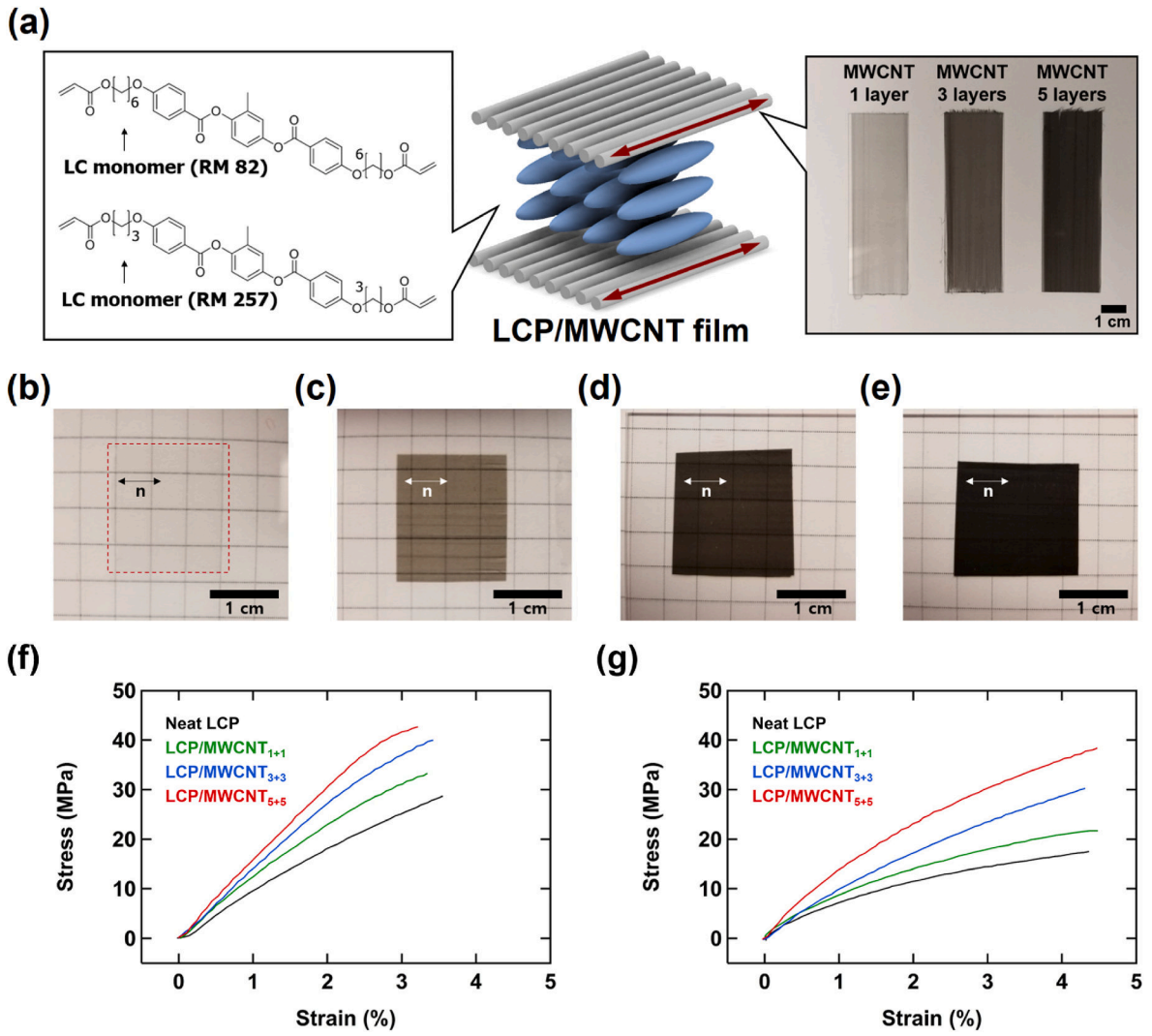



**Fig. 7.** Fabrication and mechanical properties of LCP/MWCNT composite films. (a) Molecular structures of the liquid crystal monomers (left) and schematic of the planar aligned LCP/MWCNT composite films (center); the right panel shows an optical image of layered MWCNT sheets (right). Optical images of (b) neat LCP, (c) LCP/MWCNT$_{1+1}$, (d) LCP/MWCNT$_{3+3}$, and (e) LCP/MWCNT$_{5+5}$ composite films. Nematic directors are denoted as "n." The red box in (b) indicates the location of the film on the support. Representative stress–strain curves of the LCP and LCP/MWCNT composites for (f) axial direction and (g) transverse direction to the alignment. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

exhibit improved mechanical response compared to the neat LCP. Simultaneously, the LCPs arranged concentrically around the MWCNTs efficiently transfer the external load to the outermost nanotubes because of their high interfacial stability. These effects become more significant as the concentration of the MWCNT increases and the LCP region with high crystallinity governs the nanocomposite, leading to an exponential increase in elastic stiffness. In the case of MWCNTs, load transfer is most effective at the outermost nanotube, and it drops sharply upon moving toward the inner nanotube region. This tendency is also observed in the results of strain energy per constituent atom shown in Fig. S6. The energy magnitude of the inner wall showed a reduction of more than 50% compared to that of the adjacent outer wall. This is because the interwall load transfer entirely relies on weak van der Waals interactions. As nanotubes with a smaller diameter have higher radial buckling strength, most of the strain energy is transferred to the outer nanotubes with lower strength. Therefore, the findings of this study agree with the insights based on mechanics.

To validate the simulation results, films of neat LCP and LCP/MWCNT composites were prepared as shown in Fig. 7(a)–(e). The details of the fabrication process and analysis of the composite films are provided in the experimental section in the Supplementary Data. The volume fraction of the filler was adjusted by varying the number of MWCNT

**Table 3**
Mechanical properties of neat LCP and planar aligned LCP/MWCNT composite films.

| Samples /Properties | Elastic modulus (MPa) | | Toughness (MPa) | | Failure strain (%) | | Ultimate strength (MPa) | |
|---|---|---|---|---|---|---|---|---|
| | Axial | Transverse | Axial | Transverse | Axial | Transverse | Axial | Transverse |
| Neat LCP | 1027.0 | 730.6 | 0.54 | 0.53 | 3.4 | 4.2 | 30.7 | 18.3 |
| LCP/MWCNT$_{1+1}$ | 1255.6 | 827.8 | 0.64 | 0.61 | 3.5 | 4.4 | 32.3 | 21.3 |
| LCP/MWCNT$_{3+3}$ | 1499.7 | 956.1 | 0.91 | 0.79 | 3.3 | 4.5 | 40.7 | 29.3 |
| LCP/MWCNT$_{5+5}$ | 1635.3 | 1058.9 | 1.07 | 0.94 | 3.5 | 4.4 | 45.3 | 34.7 |

layers, and the resulting composites containing 1, 3, and 5 layers of MWCNT sheet are denoted as LCP/MWCNT$_{1+1}$, LCP/MWCNT$_{3+3}$, and LCP/MWCNT$_{5+5}$, respectively. Note that as mentioned in Supplementary Data, a single layer of MWCNT sheet shows a 27% volume fraction of filler at the composite surface. Since the highly aligned MWCNT sheets were pre-placed in the mold, the MWCNTs are embedded in the matrix while maintaining a high volume fraction (see Fig. S7 in the Supplementary Data). During the subsequent interaction between the liquid crystal molecules and MWCNTs, the monomers align in parallel to the direction of the nanotubes and form a nematic monodomain across the whole system, as predicted by the simulation model (see Fig. S8 in the Supplementary Data). Especially, as the number of MWCNT sheets increases, the LCP/MWCNT composites shows better alignment features (see Fig. S9 in the Supplementary Data). The calculated orientational order parameters were 0.55, 0.58, 0.59, 0.65 for LCP, LCP/MWCNT$_{1+1}$, LCP/MWCNT$_{3+3}$, and LCP/MWCNT$_{5+5}$, respectively, which are comparable to the orientaional order of the mesogen units at the LCP-MWCNT interface as shown in Fig. 4(a). Fig. 7(f), **(g),** and Table 3 present the stress–strain curves and evaluated mechanical properties of the neat LCP sample and LCP/MWCNT nanocomposites. The elastic moduli of the LCP/MWCNT composites were significantly higher than that of the neat sample even in the transverse direction, despite the strong orientation of the MWCNTs in a specific direction. The increase in the elastic modulus was proportional to the number of MWCNT layers, resulting in an improvement of ~1.59- and ~1.45-fold in the axial and transverse directions, respectively. The fracture toughness and ultimate strength also increased with increasing MWCNT concentration in all loading directions, while the failure strain was not affected by the insertion of MWCNTs. That is, the LCP units adjacent to the nanotube surfaces had high directionality and showed efficient load transfer both in axial and transverse directions, which is consistent with the results of MD simulations.

## 4. Conclusions

In this study, the interfacial behavior of LCP nanocomposites with a high concentration of MWCNTs was examined from a nanomechanical point of view. The nanocomposite microstructures having an identical MWCNT filler geometry, but different volume fractions were modeled using all-atom MD simulations. The structural properties of the interface formed in the LCP/MWCNT nanocomposites under non-dilute filler conditions and the corresponding mechanical behaviors were investigated.

The MD simulations indicated that the equilibrated interfacial structure has a high degree of orientational order owing to the interplay between MWCNTs and adjacent LCPs. The noncovalent interactions at the LCP-MWCNT interface caused the aromatic rings of the mesogen units to repeat radially with parallel and vertical arrangements, and the effect gradually diminished over long distance. Further, the reinforcement of the elastic properties by MWCNT insertion was also predicted through uniaxial tensile tests. The results clearly suggested that the strengthening effect was exerted not only along the longitudinal direction of MWCNTs but also in transverse direction. This is another important clue that indicates the presence of strong interaction forces between the nanotube surface and LCP molecules. In this regard, we classified the mechanical contributions of each MWCNT filler and LCP matrix inside the nanocomposite by quantifying the strain energy density assigned to each component. The analysis revealed the origin of the improvement in the mechanical properties of the system with increasing volume fraction of the MWCNT, especially at high concentrations. An exponential increase in the load transfer efficiency in the interfacial region under transverse loading was also demonstrated.

Based on the findings of this study, three major conclusions were drawn. First, when MWCNTs assist better alignment of the adjacent matrix components, the liquid crystal members strongly bond to each other, resulting in improved mechanical response than that of the neat LCP system. Second, during transverse loading, LCPs effectively transfer the load to the MWCNTs with the aid of their radial alignment. Most of the external load is transformed into the strain energy of the outermost nanotube, while the amount of deformation energy decreases dramatically as the nanotubes are located deeper in the structure. Finally, this effect becomes more pronounced as the volume fraction of the MWCNT increases because the effective area with optimal radial alignment achieved by the interaction of the LCPs with neighboring MWCNTs increases. The mechanisms discussed so far explain how the microscopic interactions between co-oriented matrix and filler determine the macroscopic properties of the final product from a mechanical perspective, especially at high filler concentrations. Therefore, these findings can serve as guidelines for the design of highly loaded polymer nanocomposites, such as MWCNT forests infiltrated with LCPs, and their applications.

## Author contributions

***Hongdeok Kim***: Methodology, Software, Visualization, Writing-Original Draft. ***Hyun Kim***: Resources, Validation. ***Joonmyung Choi***: Conceptualization, Supervision, Writing-Review & Editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

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